

TECHNICAL PROPOSAL v3

Satellite-based AI Environmental Monitoring

CIMSA Cement Manufacturing Plant, Mersin

Submitted by

Airqoon (Lead) | Istanbul University-Cerrahpasa, Dept. of Environmental Engineering (Academic Partner)

Duration

12 months (M) · WP 1: M1–3 · WP 2: M3–6 (parallel with WP 3) · WP 3: M3–11 · WP 4: M10–12

SECTION A — Methodology & Approach to the Assignment

1. Program Overview

Airqoon, in partnership with the Department of Environmental Engineering at Istanbul University-Cerrahpasa, proposes a 12-month, four-step programme to deliver the complete Dust Management Plan and all TOR deliverables within a one-year period. The programme is structured to reflect the natural progression of the work: historical baseline and desktop review, dust source mapping and site assessment running in parallel with modelling (~~both plume and~~ particle dispersion modelling) and monitoring system design, concluding with the consolidated DMP and Final Environmental Impact Report. All six TOR tasks are addressed, and all six TOR deliverables are produced. The programme is conducted in full alignment with EBRD ESR3, EU IED 2010/75/EU, the BAT Conclusions for Cement/Lime/Magnesium Oxide, EU CAFE Directive 2024/2881, and WHO Air Quality Guidelines 2021.

2. Four Work Package Programme (12 months)

Work Package 1 (WP 1)—Desktop Review, Data Collection & Historical Baseline [Months 1–3] / TOR T1, T4

This work package establishes the pre-investment air quality reference conditions and produces the historical baseline against which all subsequent monitoring and compliance demonstrations will be measured.

Regulatory & operational document review (TOR T1)

This TOR aims to assess the current situation of CIMSA regarding emissions and its effect on ambient air quality management

- Review of all CIMSA Mersin environmental permits, stack emission reports, dust emission inventory, flue gas monitoring data, existing AERMOD dispersion model outputs, ambient air monitoring and operational logs.
- A comprehensive gap analysis will be performed to evaluate current operations against the Best Available Techniques (BAT) Conclusions for the production of Cement, Lime, and Magnesium Oxide. This evaluation focuses on aligning site practices with established European standards for industrial emissions control and environmental management. BAT 5 (Monitoring): Evaluation of the existing maintenance management systems for both continuous and periodic measurement of dust (particulate matter), NO_x, SO_x, and CO emissions (combustion emissions) across all kiln and non-kiln sources. BAT 14–15 (Diffuse Dust): Assessment of diffuse and fugitive emission sources, including open material stockpiles, conveyor systems, and quarry haul roads, against required controls such as enclosure, covered transport, and road wetting. BAT 16–18 (Channelled and Process Dust): Review of point-source emissions from kiln stacks, cooling units, and milling operations to ensure the effectiveness of dust control equipment like filters and electrostatic precipitators.
- Regulatory review: Türkiye's Regulation on Industrial Emission Management (resmi gazete:32782, 2025), Industrial Air Pollution Control (27277, 2009); EU IED 2010/75/EU Annex I §3.1; EU CAFE Directive 2024/2881 Annex I limit values and 2030 targets.

Historical air quality data collection (TOR T4)

This TOR aims to determine long-term and spatial ambient air pollution (for aerosol) around the CIMSA using satellite and ground-based air quality monitoring systems.

- Satellite-derived AOD: MODIS MAIAC (500 m, daily); Sentinel-5P/TROPOMI (NO₂, SO₂, CO columns).
- Ambient air pollution data from the air quality monitoring stations (AQMS) close to the CIMSA operated by the Environment, Urban and Climate Ministry and operated by CIMSA (if available). All monitoring parameters (such criteria pollutants: PM₁₀/PM_{2.5}, SO₂, NO_x, CO, O₃) will be evaluated.

- Satellite-derived data: MODIS MAIAC AOD (500 m, daily) will be used as an indicator of atmospheric aerosol loading, while Sentinel-5P/TROPOMI data (NO₂, SO₂, and CO column densities) will be used to support the assessment of gaseous air pollution around the plant.
- Stack emissions (SEÖS) data analysis and evaluation, comparing the CIMSA operation information (if available).
- MODIS MAIAC AOD will be used to generate spatial distribution maps of aerosol loading around the CIMSA plant and to support the interpretation of ground-based PM measurements.
- Data provided by CIMSA: stack records, dust complaints log, existing sensor or settled dust deposition data, any internal dust source inventories, including environmental aspects tables prepared under ISO 14001.
- ERA5/MERRA-2 meteorological reanalysis: hourly wind, mixing height, stability class that are inputs for dispersion modelling and baseline meteorological/air quality characterization.

Output

- Baseline Environmental Report (TOR Deliverable 1), establishing air quality reference levels with particular emphasis on PM₁₀ and PM_{2.5} exceedance events and their frequency.
- Initial KPIs for future monitoring; identification of data gaps; and explicit recommendations for the scope and configuration of the detailed dispersion modelling (WP 3) and the long-term continuous monitoring system (WP 2), ensuring a documented and auditable link between the baseline findings and all subsequent monitoring design decisions.

Work Package 2 (WP 2) — Dust Emission Mapping, Site Visit & Monitoring System Design, Source Apportionment [Months 3–6] / TOR T2, T3, T5

This is the core field phase. A dedicated dust source mapping exercise and a structured site visit that together produce the ground-truth evidence base for the dispersion model and DMP. In parallel, we design, specify, and commission the long-term monitoring infrastructure, a sensor network, physicochemical sampling programme, and dashboard, as fully operational tools handed over to CIMSA for ongoing use after the assignment concludes.

Dust emission mapping and receptor identification (TOR T2)

Drawing on the desktop review outputs, we will compile a georeferenced emission source inventory covering the full area of influence — not only the plant boundary but also quarries, access roads, and the Tekke/Yaka receptor zone.

- Dust type categorization per TOR taxonomy: (i) Raw Material Dust: crushing, grinding and handling of limestone, clay, and correctives; primarily coarse particles, (ii) Kiln and Clinker Process Dust: calcination, kiln firing and clinker cooling; fine reactive particles i.e. alkali metals, and trace heavy metals, (iii) Cement Dust: clinker grinding, milling, storage, and packing, very fine Portland cement minerals with gypsum and additives. Each category requires distinct monitoring techniques and control measures addressed separately throughout the inventory and DMP.
- Point/channeled sources: kiln stack, raw mill, clinker cooler, cement mill, packaging lines, each assessed against BAT-AEL thresholds (BAT 16–18) and current dedusting equipment (ESP/fabric filter/hybrid filter per BAT 17–18).
- Diffuse/fugitive sources: open material stockpiles, raw material conveyors and elevators, crusher feeds, clinker storage, dispatch loading bays, catalogued against BAT 14 technique list (enclosure, covered conveyors, negative pressure ventilation, closed silos with fabric filters).
- Road and quarry dust: haul roads within the plant, external access roads to limestone quarries, assessed against BAT 15 controls (paving, road wetting, wind barriers, stockpile humidification).
- Sensitive receptors: Tekke and Yaka settlements, agricultural land, any schools, healthcare facilities, or other sensitive uses within the area of influence, mapped with distance-weighted exposure classification.
- Current operational controls assessment: Gap analysis between existing control measures and BAT 14–18 requirements. Preliminary improvement proposals and upgrade recommendations will be included in the Dust Source Map & Receptor Report (Deliverable 2), providing an actionable interim output ahead of the full DMP control hierarchy developed in WP 4.

Site visit and field verification (TOR T3) — Month 3

Conducted by Barış Can Üstündağ (Airqoon Technical Lead) and Dr. Coşkun Ayvaz (Istanbul University-Cerrahpasa), the site visit is a standalone, structured field assessment with the following objectives:

- Systematic walkthrough of all production areas (quarry face, crusher, raw material storage, raw mill, kiln, cooler, cement mill, packing, dispatch) to observe and document visible dust emissions, spillage points, and material handling conditions.
- Assessment of operational controls in practice: Questions planned to be asked: Are BAT 14–15 enclosure, suppression, and housekeeping measures functioning as intended? Are filters and ESPs maintained in accordance with BAT 16?
- Structured interviews with CIMSA team: Environmental manager, H&S manager, and operations supervisors to capture: incident history, community complaint records, maintenance schedules, and any planned control investments.
- Identification of issues not visible in desktop review: Road dust from internal haul traffic, wind-blown dust from exposed stockpiles, dust deposition patterns on surrounding vegetation, and any storage or handling practices creating uncharacterized emission points.
- Finalisation of sensor node placement: confirmation of the *6 particulate matter sensor node* locations based on dispersion model preliminary outputs and direct observation of prevailing dust pathways.

- Photographic and drone-survey documentation of the site perimeter, receptor settlements, and key dust source locations.

Monitoring system design and sensor network specification (TOR T5)

- The monitoring system design draws directly on Airqoon's existing operational deployments at cement facilities, where a comparable sensor network, alarm system, and AI-backed monthly reporting module are already in production use.
- We specify and deploy a *6-node Airqoon sensor network: 2 × Unit L-PRO (24-bin PNSD, 0.35–40 µm) at modelling-critical positions, and 4 × Unit L (4-bin OPC, PM1/PM2.5/PM10) covering Tekke/Yaka receptor zones, quarry access road, and kiln-proximate fence-line.* The network spans the full perimeter including quarry access roads and Tekke/Yaka receptor zones, rectifying the directional coverage gap of existing monitoring stations. The network is commissioned during this assignment and fully handed over to CIMSAs as an operational asset for long-term continuous monitoring beyond the contract period.
- QA/QC by Gülkan Güner: By following the draft standard of CEN/TS 17660-2, 4-week co-location validation will be conducted against reference-grade (gravimetric measurement) monitor; calibration certificates and full QA documentation handed to CIMSAs for ongoing self-management.
- Cloud ingestion pipeline, real-time dashboard, and automated monthly periodic reports (Orkun Şentürk), commissioned and handed over by Month 6.
- Airqoon operates an existing cloud data infrastructure for environmental sensor network management, real-time data streaming, and visualization, which is developed and battle-tested across prior deployments.

Physicochemical sampling programme design and initiation

- *CIMSAs will provide chemical and physical characterisation results for particulate matter at selected locations around the plant by CIMSAs and project manager. The chemical and physical characterisation of particles will allow us to determine the source contribution of this location: such as combustion, process emission, traffic dust transport etc. via modelling the data by positive matrix factorisation (EPA-PMF Model). This source characterisation shows the primary contribution of sources. The consortium's independent filter-based sampling programme therefore focuses specifically on ambient receptor locations i.e. perimeter fence-line and the Tekke/Yaka settlements to characterise the mixed signal at receptor points and validate source attribution model outputs. Sampling follows an every 6th day protocol over Months 3–10, yielding approximately 35 sampling events and ~70 PM_{1.0}/PM_{2.5} filter pairs for gravimetric mass, metal, ion, and OC/EC analysis. The sampling protocol and laboratory partnership are operational by Month 6 and handed over to CIMSAs for independent continuation.*
- *CIMSAs will provide chemical and physical characterisation results for particulate matter at selected near-source locations around the plant, covering gravimetric mass, trace metals, water-soluble ions, and OC/EC fractions. This characterisation establishes the plant's emission fingerprint across source types — cement process emissions, combustion, and traffic dust transport — and provides the primary source profile inputs required for EPA-PMF receptor modelling. Building on this foundation, CIMSAs further provides a PM10 filter sampler, deploys and operates it at two receptor locations (Tekke and Yaka settlements), and performs all filter analysis — gravimetric mass, trace metals, water-soluble ions, and OC/EC — in its in-house laboratory, in accordance with TOR §8. One hundred (100) PM10 filter samples are collected over the sampling period — fifty (50) at Tekke and fifty (50) at Yaka. The consortium provides the sampling protocol, chain-of-custody documentation, and full EPA-PMF source apportionment analysis on the combined near-source and receptor-point dataset, resolving contributions from cement process emissions, combustion, and background transport sources; results are reported in the Source Apportionment Report (D-5).*

Output

- Dust Source Map & Receptor Report (TOR Deliverable 2), including a comprehensive georeferenced emission source inventory and preliminary improvement proposals against BAT 14–18 requirements.
- Site Visit Report (TOR Deliverable 3), containing field verification findings, interviews, photographic, and drone-survey documentation.
- Monitoring System Specification & Handover Package (TOR Deliverable 5), fully specifying the air sensor network, QA/QC protocols, and the cloud ingestion pipeline.
- Fully operational and handed-over Real-Time Monitoring Dashboard and the automated system for Monthly Periodic Reports (TOR Deliverable 4).
- *Physicochemical Sampling Protocol and PMF Analysis Framework, delivered to CIMSAs by Month 4 to govern the filter sampling campaign and source apportionment analysis.*
- Source profile report.

Work Package 3 — Dispersion Modelling, AI/ML Analytics & Dashboard [Months 3–10, parallel with WP 2] / TOR T5

Running in parallel with field activities from Month 3, WP 3 builds the analytical intelligence layer that underpins the DMP, satisfies TOR Deliverable 4, and provides CIMSAs and EBRD with quantified, evidence-based emission characterization.

Atmospheric modelling suite

Rather than relying on a single dispersion model, we develop an integrated modelling suite covering four complementary capabilities, led by Prof. Ülkü Alver Şahin and Barış Can Üstündağ on model integration and data pipelines:

- Particle size distribution modelling: Estimating how PM₁₀ and PM_{2.5} concentrations distribute spatially under varying atmospheric and meteorological conditions, producing the visual representations of PM₁₀ and PM_{2.5} distribution patterns required by TOR Deliverable 4. ERA5 meteorological reanalysis provides the atmospheric inputs; PNSD measurements from the sensor network provide ground-truth size distribution data for model calibration and validation.
- *Plume modelling: Simulating the shape, vertical extent, and horizontal spread of emission plumes from individual point sources (kiln stack, clinker cooler, cement mill) under different wind speed, atmospheric stability, and temperature conditions. Outputs identify the conditions under which specific sources pose the greatest receptor risk.*
- Particle dispersion modelling: Tracking how particles of different size fractions travel and deposit across the area of influence. Coarser particles (PM_{2.5}-PM₁₀) settle within the plant boundary; fine (PM_{2.5}) fractions travel further and govern off-site receptor exposure. This directly informs the sensitive receptor analysis in WP 4.
- Trajectory analysis: Forward and backward trajectory modelling by HYSPLIT to establish the transport pathways between emission sources and receptors, identifying which sources contribute to air quality at Tekke and Yaka under prevailing wind conditions, and under which meteorological scenarios receptor concentrations approach or exceed limit values.

Satellite-based time-series plume quantification

Sedef Bayram with contributions from Barış Can Üstündağ and Coskun Ayvaz PhD. leads a systematic analysis of time-series Sentinel-2 and MODIS imagery to quantify dust plumes to estimate plume extent, frequency, and relative intensity over the 2018–present archive. Image processing algorithms are applied to measure dust plume area and optical depth as a function of time, season, and operational activity, producing the timelapse analysis required by TOR Deliverable 2.

Multi-source ML fusion model for source detection and attribution

Barış Can Üstündağ contribution from Sedef Bayram, leads the development of an ML model that fuses four data streams simultaneously:

- *PM Dispersion Maps & ML Source Attribution Report (TOR Deliverable 4), providing quantified evidence of emission contribution and impact.*
- Satellite imagery (Sentinel-2 multi-spectral, MODIS AOD)
- Ground sensor data (OPC for PNSD and PM mass concentrations from the *6 node network*)
- Atmospheric and meteorological data (ERA5 wind fields, field meteorological measurements, mixing height, stability class, precipitation)
- CIMSA manufacturing process data (kiln feed rates, production schedules, operational logs which is provided by CIMSA per TOR §8 obligations)

Machine learning models process all four streams simultaneously to identify pollution sources, quantify their relative contributions to receptor-level PM concentrations, and detect trends and anomalies over time. The model outputs feed directly into the monitoring dashboard and into the source attribution section of the Final DMP.

Integrated monitoring dashboard

Orkun Şentürk builds and hands over the *Airqoon* cloud platform by Month 6: Aggregating all four data streams, displaying real-time PM trends and modelling outputs, generating KPI compliance scorecards against CAFE 2024/2881 and BAT-AEL thresholds, and producing automated monthly reports. *The fully configured platform, all underlying models, data pipelines, and user documentation are formally handed over to CIMSA as a permanent operational asset for independent continuation of the monitoring programme. Full operational handover to CIMSA with user documentation.*

Output

- PM Dispersion Maps & ML Source Attribution Report (TOR Deliverable 4), providing quantified evidence of emission contribution and impact.
- Timelapse Change Detection Report (TOR Deliverable 2), based on 2018-to-present satellite imagery analysis.
- Initial Physicochemical Dust Report (TOR Deliverable 8), including initial filter results and the full programme specification.
- Quantified, evidence-based emission characterization, including annual average PM concentration maps and exceedance frequency maps, derived from the integrated atmospheric modelling suite.
- *Airqoon Lens platform instance, fully configured and operational, with complete user documentation (TOR Deliverable 6).*

Work Package 4 (WP 4) — Dust Management Plan & Final Reporting [Months 10–12] / TOR T6

Integrating all evidence from WPs 1–3, baseline data, source inventory, site visit findings, dispersion model outputs, ML source attribution, and initial physicochemical results. WP 4 delivers the complete, ESAP-compliant Dust Management Plan and Final Environmental Impact Report by Month 12, fully satisfying the EBRD one-year development commitment.

Dust Management Plan — required components (TOR T6)

- Emission inventory: Source-specific measurements and estimates for all channeled, diffuse, and fugitive sources, benchmarked against BAT-AELs (BAT 16, Dusty operations; BAT 17–18, kiln/cooling/milling).
- Process-specific dust control matrix: hierarchy of controls per dust category (raw material, kiln-clinker, cement grinding/packing) referencing BAT 14–16 techniques: enclosure, covered conveyors, closed storage silos with fabric filters, negative-pressure ventilation, water spray, road wetting, paving.
- Sensitive receptor analysis: Distance-weighted PM exposure assessment for Tekke and Yaka, with health risk context against WHO AQGs 2021 and EU CAFE 2024/2881 limit values.
- Operational improvements and control hierarchy: Ranked recommendations against BAT 14–18, with capital cost estimates where available and prioritized implementation sequence.
- Worker protection and training: Occupational exposure limit compliance roadmap; dust control awareness training programme for kiln, clinker handling, and dispatch personnel; PPE requirements aligned with BAT general principles (BAT 1, CLM BREF §4.1.1).
- Monitoring strategy and KPIs: PM_{2.5} and PM₁₀ annual mean (CAFE 2024/2881 Art. 9); BAT-AEL stack compliance; settled dust deposition rates at receptor locations; community dust complaint frequency index.
- Grievance and complaint management: Structured protocol for logging, investigating, and responding to dust complaints from Tekke and Yaka communities; integration with CIMSA's Stakeholder Engagement Plan and Community Health and Safety Management Plan as required by the ESAP.
- Implementation roadmap with phased actions, responsibilities, and a timeline for implementation and verification including interim verification checkpoints, performance review dates, and the schedule for demonstrating progressive alignment of CIMSA operations with 2030 EU ambient air quality targets (EU CAFE 2024/2881) and WHO Air Quality Guidelines 2021. This timeline is structured to support EBRD's ongoing ESAP progress reporting obligations.

Output

- Final Dust Management Plan + Environmental Impact Report (TOR Deliverable 5), delivering the complete, ESAP-compliant document.
- Full suite of required components: Emission Inventory (benchmarked against BAT-AELs), Process-specific Dust Control Matrix, Sensitive Receptor Analysis, Implementation Roadmap, and Grievance/Complaint Management Protocol.

4. Assignment Milestones — 12-Month Timeline

Activity / Month →	Q1	2	3	Q2	5	6	Q3	8	9	Q4	11	12
WP 1: Desktop Review & Baseline												
Regulatory/doc review + gap analysis												
Satellite/AQMS/SEÖS data												
Baseline Environmental Report •			•									
WP 2: Dust Emission Mapping, Site & System												
Dust emission mapping + receptor ID (T2)												
Site visit & field verification •			•									
Air sensor network deployment + QA/QC •						•						
Dashboard build + handover •						•						
Receptor filter sampling (perimeter+receptors)•				•								
Physicochemical Dust Report •											•	
Monthly periodic reports •				•		•			•			•
WP 3: Modelling, AI & Dashboard												
Atmospheric modelling suite												
ML source detection + attribution models												
Timelapse change detection report •								•				

Remote Sensing Technologies

- Sedef Bayram holds an MSc and pursuing a PhD, specialising in remote sensing and satellite imagery analysis, with hands-on expertise in multi-spectral image processing, satellite-derived AOD retrieval (MODIS MAIAC, Sentinel-5P/TROPOMI), and change detection that is directly applicable to the satellite-based plume identification required by this assignment. She is working in IUC-APRG as a research assistant.
- The consortium routinely works with the satellite sensor constellation: Sentinel-2 (optical/NIR, 10 m), Sentinel-5P/TROPOMI (UV-visible spectrometry for NO₂/SO₂/CO), MODIS (thermal infrared, aerosol optical depth), and CALIPSO (satellite-based LIDAR aerosol profiling). For this assignment, ground-based in-situ LIDAR is replaced by PM sensor PNSD size-resolved measurements and satellite CALIPSO profiles, providing equivalent vertical dust characterisation within the available project scope.

Air Quality Dispersion Modelling

- Prof. Şahin's research interests centre on dispersion and receptor modelling of air pollutants, and she has published in Atmospheric Chemistry and Physics on wide-range particulate matter size distribution interpretation, including a measurement report from Delhi co-authored with the University of Birmingham Atmospheric Research Group, where she conducted research in 2019–2020. This international experience in industrial and megacity atmospheric environments directly supports the modelling suite proposed in WP 3.
- Coşkun Ayvaz PhD supports model input preparation, emission inventory development, and meteorological data processing, directly applicable to AERMOD input preparation for the Mersin site. He will also PM monitoring and sensor measurements, supporting the generation of robust field data for model development, calibration and interpretation. In addition, he has contributed to the Cebeci mining environmental plan for a dust control project focused on dust control, providing practical experience directly relevant to DMP.

EU & National Environmental Regulations

- The consortium has working familiarity with EU Directive 2024/2881 (CAFE recast), EU IED 2010/75/EU, BAT Conclusions for Cement/Lime/Magnesium Oxide (Commission Decision 2013/163/EU, specifically BAT 5 monitoring, BAT 14–18 dust), and Türkiye's Regulation on the Control of Air Pollution from Industrial Sources.
- Prof. Ülkü Alver Şahin teaches environmental law and regulation as part of the Istanbul University-Cerrahpasa Environmental Engineering curriculum, ensuring regulatory interpretation is embedded in all technical outputs, particularly the DMP's compliance roadmap to EU CAFE 2024/2881's 2030 limit values.

2. Data Science and AI Integration

Machine Learning & Predictive Analytics

- Barış Can Üstündağ has a peer-reviewed publication record in machine learning and computer vision, including image classification and sensor data fusion architectures. These methods transfer directly to the satellite-based dust plume detection and source attribution tasks required by this assignment, significantly reducing model development time relative to a team starting from scratch.
- Airqoon's internal ML pipeline covers: classification models for air quality anomaly detection; (directly applied by Gülkan Güner to PM sensors QA/QC); time-series forecasting for PM trend analysis; all deployed on real-world environmental sensor datasets at industrial sites.
- The consortium is an established working team, not a new arrangement for this proposal. Barış Can Üstündağ, Coşkun Ayvaz, and Prof. Şahin have co-authored peer-reviewed work together (Ayvaz et al., 2024, Spectrochimica Acta Part A Journal), and Coşkun Ayvaz is lead author on a 2025 low-cost sensor review paper supervised by Prof. Şahin (Ayvaz et al., 2025, Environmental Pollution Journal). The team also prepared a paper about a spatial range particulate sensor used in an industrial area (Inegöl/Bursa). The paper is under review in the Atmospheric Pollution Research journal.
- The team has experience deploying ML inference pipelines in cloud environments (AWS/GCP).

Custom Dashboards & Visualisation Tools

- Airqoon maintains a production cloud platform for multi-site environmental sensor data ingestion, management, and real-time visualisation, currently operational across deployments in the air quality monitoring sector. Orkun Şentürk, who designed and maintains this infrastructure, will adapt it directly for the CIMSA assignment which will bring a proven, operational system rather than a greenfield build.
 - Orkun Şentürk specialises in cloud data architecture and interactive environmental monitoring dashboards, having designed and built multi-source data ingestion systems integrating IoT sensor streams, satellite data feeds, and meteorological inputs into unified real-time visualisation platforms.
 - The consortium has delivered custom dashboard interfaces for environmental monitoring clients featuring: time-series PM trend plots, geospatial pollution maps, KPI compliance indicators, and automated PDF/email reporting.
 - All dashboard components are built on open-source technology stacks (Python, PostGIS, Grafana/Dash) ensuring long-term maintainability by CIMSA and EBRD without proprietary vendor dependency.
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3. Field Implementation

On-site Assessments & Stakeholder Engagement

- Airqoon has conducted structured site assessments at industrial facilities including cement plants, interviewing operational and H&S personnel, documenting emission sources against BAT checklists, and identifying monitoring locations through systematic walkthrough surveys.
- Barış Can Üstündağ has international field deployment experience, coordinating sensor installation across diverse regulatory and operational environments, including liaison with municipal environmental authorities, Ministry of Environment regional offices, and industrial facility environmental teams. He is familiar with the access constraints, safety protocols, permit requirements, and operational schedules typical of cement manufacturing facilities.
- Coşkun Ayvaz PhD has hands-on field experience in environmental engineering, including site surveys, sampling campaigns, and on-site data collection at industrial facilities, which is directly applicable to the source verification, filter sampling logistics, and field documentation tasks at the Mersin site and associated quarries.
- The consortium has the logistical capacity to conduct multi-day field campaigns at the Mersin facility and associated quarry sites, coordinating equipment deployment, drone survey operations, filter sample collection, and personnel interviews within CIMSA's operational schedule.

4. Communication and Reporting

Technical Writing for Lender-Grade Disclosure

- All consortium deliverables are structured for EBRD lender review: evidence-based, clearly referenced against regulatory thresholds (BAT-AELs, EU CAFE 2024/2881 limit values, WHO AQGs 2021), and written in plain English suitable for disclosure to international financial institutions and affected communities.
- Prof. Ülkü Alver Şahin has an established track record of producing peer-reviewed publications and technical reports for regulatory bodies and international funding organisations in both Turkish and English, providing scientific quality assurance over all consortium outputs.
- Airqoon produces English-language monitoring reports and technical documentation for international environmental compliance frameworks as a standard business practice.

Project Team

Name	Organisation	Role	Competency Coverage
Barış Can Üstündağ	Airqoon (Lead)	Technical Lead & ML/AI	1 (cement sensor deployment, industrial monitoring), 2 (ML/CV published, sensor fusion), 3 (field deployment, site assessment)
Gülkan Güner	Airqoon	Sensor QA/QC	1 (PM sensor calibration, reference co-location protocol), 3 (sensor deployment logistics)
Orkun Şentürk	Airqoon	Cloud & Dashboard	2 (cloud architecture, real-time dashboards, automated reporting), 4 (reporting infrastructure, open-source stack)
Prof. Dr. Ülkü Alver Şahin	IUC Environmental Engineering	Scientific Director	Experience with dispersion/receptor modelling, industrial emissions and ambient air quality and regulation, (more scientific papers, presentations, reports and projects. director of IUC Air Pollution Research Group; many international collaborations, such as with visiting researchers from the University of Birmingham, Carleton University, NOA in Athens at the Atmospheric Research Group
Research Assistant Coşkun Ayvaz PhD	IUC Environmental Engineering.	Environmental Engineer, Researcher	1 (emission inventory, regulatory review, BAT gap analysis), 3 (field surveys, Turkey stakeholder coordination , filter sample logistics, on-site data collection)
Research Assistant Sedef Bayram	IUC Environmental Engineering	Environmental Engineer, Researcher	1 (satellite remote sensing MSc — optical/NIR/TIR/spectrometry platforms; Sentinel, MODIS, CALIPSO; AOD retrieval, change detection, GIS)